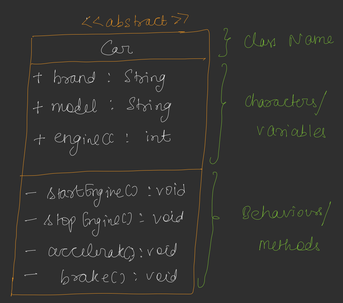
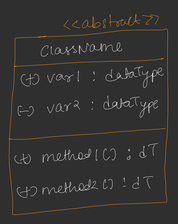
Class Diagram

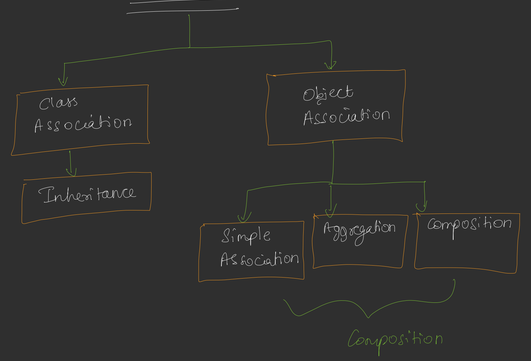
```
Class Car {
    String brand;
    String model;
    int engineCC;
    startEngine();
    stopEngine();
    accelerate();
    brake();
}
```



Generic Class



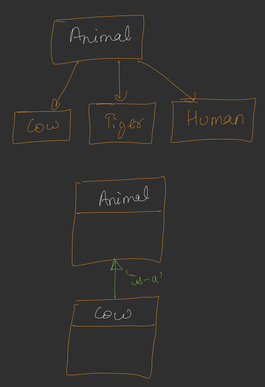
Associations



Inheritance

- (is-a) relationship
- Denoted by (⟶)

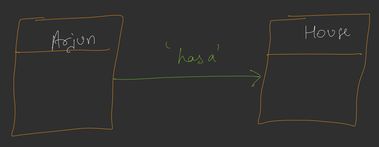
```
Class A {
    method1();
}
Class B : Public A {
    method2();
}
main() {
    B* b = new B();
    b->method1();
    b->method2();
}
```



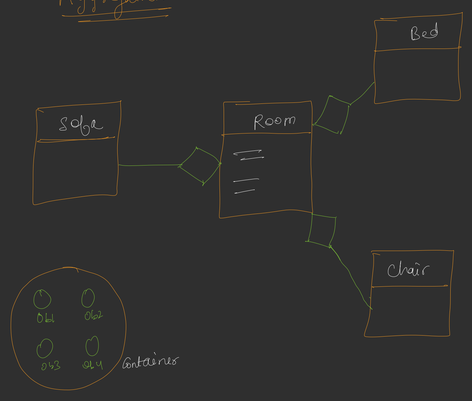
Composition

- (has-a) relationship
- Denoted by (⟶)

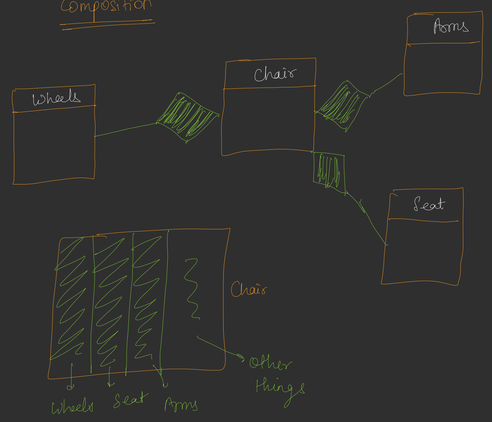
Simple Association



Aggregation



Composition



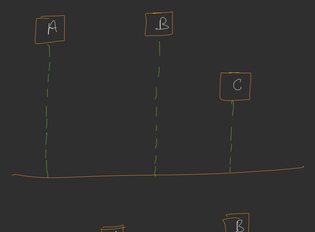
Composition in Code

```
Class A {
    method1();
}

Class B {
    A * a;
    B() {
        a = new A();
    }
    method2();
}

main() {
    B * b = new B();
    b -> method2();
    b -> a -> method1();
}
```

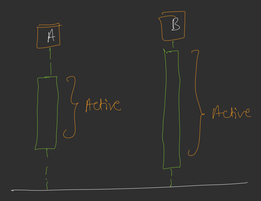
Sequence Diagrams

① Representing Object:

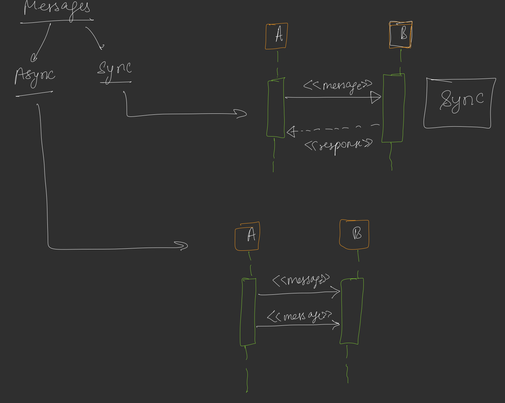
② Lifeline



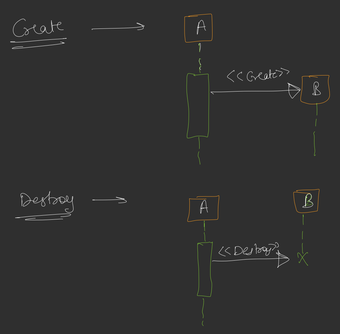
② Activation Bar



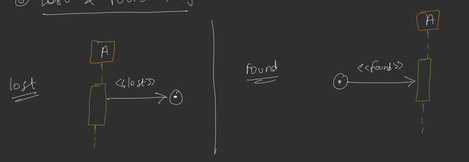
④ Messages



⑤ Create & Destroy Message



⑥ Lost & Found Msg



ATM Transaction flow

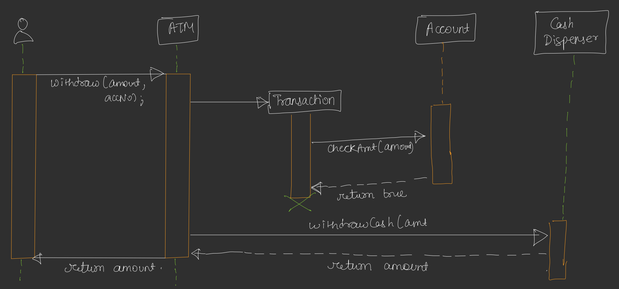
Use Case

1. → ATM
   (acc, amount)
2. ATM → Transaction

→
1. Verify ATM pin
2. verify Account
3. Tell Cash Dispenser for money.

Sequence Diagram

1. Use Case
2. Objects
   - ATM
   - User
   - Transaction
   - Account
   - Cash Dispenser
3. Draw Diagram



alt : (if-else)

Option : (if)

loop : for / while